AI-Augmented Systems Engineering

The 'Audit & Simulate' Framework for C++ Development

Part 1: The Master Audit Protocol

Anchoring complexity through technical guardrails.

Defining the Protocol

The Technical Anchor

An Audit Protocol is a persistent "Source of Truth" that prevents the AI from losing track of code modularity and narrative depth over long development sessions.

Enforcing Standards

By defining a "Master Build Status," we ensure that every surgical fix (e.g., Collision Detection) respects all previous structural requirements of the 5-module system.

Locking Lore & Logic

Prompt Example:

```
"[MASTER AUDIT]: Ensure Phase 3 high-fidelity
descriptions remain. Restore the Amalgamation Boss
logic. Verify 0 build errors in Dungeon_phase.exe."
```

This command forces a comprehensive regression test across all modules before any code output is generated.

The Three Pillars of Auditing

Consistency

Maintaining a unified variable namespace across five independent executables.

Integrity

Preventing the deletion of complex 'Switch-Case' lore descriptions during optimization.

Safety

Ensuring cross-process string conversions (std::string to c_str) are always applied.

Part 2: Logical Simulation

Using the AI as a real-time sandbox processor.

Simulation vs. Compilation

100%

Logic Verification

The Sandbox Processor

While a compiler checks for Syntax, Gemini simulates Physics. We test movement, coordinate updates, and boss probability AI directly in the chat window, catching bugs that a compiler would ignore.

Case Study: The 'Ghosting' Bug

Bug Finding via Demo

During a simulation of the Main Hall, I attempted to walk 'Down' into a wall. The AI initially let the marker [@] overlap the wall ###.

Result: This simulation finding led us to implement the 'Collision Shield' logic gate in the Mansion module.

Simulation-Led Resolutions

Scenario Tested	Observation (Bug)	Resolution (Fix)
Wall Approach	Player icon overwrote wall icon	Inserted 'if(next==0) continue;' gate
Room Exit	Player reset to room center	Implemented Smart-Door memory
Alchemical Trial	Boss skipped scrap-turns	Refined probability AI (S > E)

Monday Demo Roadmap

- **Initialize State:** Feed Gemini the current Mansion coordinates (3,1).
- **Request Map:** Ask for a high-fidelity ASCII render of the Main Hall.
- **Test Physics:** Walk directly into a wall to demonstrate the Collision Shield.
- **Trigger Handoff:** Walk onto the [K] tile to show the cross-module transition logic.

"Debugging is not just about finding errors in syntax; it is about simulating the truth until the error becomes visible."

— Project Thesis: AI as Logical Sandbox

Questions?

Final Build: LOCKED [0 Errors | 0 Warnings]

www.github.com/TheOozingCrown | May 8, 2026

Image Sources

https://static.vecteezy.com/system/resources/previews/065/655/176/non_2x/green-color-security-shield-icons-for-protection-safety-themes-vector.jpg

Source: www.vecteezy.com

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https://png.pngtree.com/png-vector/20221116/ourmid/pngtree-red-location-map-icon-pin-navigation-pointer-target-png-image_6460378.png

Source: pngtree.com